VF Asia Footwear Laboratory Testing Manual

Test Method for Water Penetration: Maeser Test	Test Method #: VF- E11 REF #:TM-R- 022 Page 1 of 3
Date Issued: November 27, 2018 Revision #: 3 Reason for revision: Corrected the right TMR Ref #	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To determine the resistance of material to water penetration on flexing.

REFERENCE: ASTM D 2209 /SATRA TM 34

EQUIPMENT AND MATERIAL:

1 Maeser machine which has pairs of V-shaped clamps
2 A press knife or cutting device.
3 Tap water ***(Distilled or de-ionized water to resolve lab discrepancy)***
4 Thin strips of compressible impermeable material, such as soft rubber or plastic.
5 Rubber sealant or similar type of compound.
6 Leather, fabric

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at standard laboratory environment at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.
2. Die or hand cut a test sample which is 114 ± 2 mm by 114 ± 2 mm so that one side of each specimen is parallel to the principal direction of the material. If possible, take the specimens from position across the full usable width and length of the material. For a material with a woven structure this should ensure that the specimens do not contain the same warp or weft threads.
3. Mark the principal direction of the material on each test specimen.
4. If it is considered that the test specimens are virtually impermeable through their thickness, but may be prone to wicking along the length, then seal all four edges of both test specimens with the rubber sealant.

   Note: For sample intended for "Dynamic Absorption Test ", weight each test piece. W1

REPETITIONS: Test two (2) samples per material**.**

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TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Adjust the test machine so that a pair of clamps and have their tips at their maximum separation and in the same horizontal plane.
2. If it is considered that the test specimens are virtually impermeable, through their thickness but may be prone to wicking along their length, then place a thin strip of compressible impermeable material in each of the clamps in the area where the test specimen will be bedded down. This will help to prevent water seeping between the test specimen and the clamps during the test.
3. Fold, without creasing, one test specimen in half so that the outer surface is facing outwards and the fold is parallel to the principal direction of the material.
4. Place the folded test specimen between the pair of clamps so that the fold runs between the tips of the “V”s.
5. Fully tightened one of clamps, ensure that the test specimen is not slack, and then fully tightened the other clamp.
6. Slowly move the clamps together and watch the specimen to ensure that the center section folds upwards. If this is not the case gently apply pressure to the underside at the center of the fold as the clamps move together to form an upward fold.
7. If the test machine has second pair of clamps, repeat the procedure for the other test specimen but this time fold the specimen at 90 ° to the principal direction of the material.
8. Fill the container with water and adjust the level so that it is above the center of the upward fold. During this stage it is recommended that a wad of absorbent tissue is put in the trough formed by clamped test specimen. This wad should be removed from the specimen after water level has been adjusted.
9. Immediately start the test machine. Inspect the interior of the clamped test specimen(s) for signs of water penetration. It is usual for this to occur initially at the two ends of the center fold and take the form of a damp patch at the surface of the material, or a globule of water exuding from it.

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10. If water starts to seep between the test specimen(s) and the clamps, this is invalid and should be discounted and the test continued until valid penetration through the test specimen occurs. If the amount of seepage through the clamps is enough to put the test at risk stop the machine and mop it up using an absorbent tissue. Check the clamps and if necessary tighten them further.

11. At first sign of valid water penetration through the test specimen record the number of cycles completed by the movable clamp.

12. Continue to inspect the test specimens for approximately 15 minutes or until the test specimen (s) show signs of initial water penetration. Do not stop the machine when making the inspection.

13. If penetration has not occurred after approximately 15 minutes, then repeat the procedure but gradually increasing the inspection intervals from every few minutes to every quarter of an hour or more should the material continue to resist penetration. When penetration occurs during a period of intermittent inspections, record both the number of cycles of the last inspection stage before penetration and the first inspection stage after penetration.

14. Continue the test until the required number of flexes.

15. **To determine water absorption percentage**, remove the leather cutting from the machine, blot the excess surface water with absorbent paper, and weigh each cutting to the nearest 0.01g. **W2**: ***NOTE: When comparing water absorption values, all cutting specimens should receive the same number of flexes.***

CALCULATION:

Water absorption % = [(W2 – W1) / W1] x 100

W1 = weight of specimen before flexing

W2 = weight of specimen after flexing

REPORT:

1. Maeser Test: Number of cycles when penetration occurs

2. Dynamic Water Absorption: %